

Four in five vertebrate IP₃ receptor genes cannot be reached from any protein database

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Running title: How the IP₃ receptor genes are recorded

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Abstract

Gene-family studies find their genes through protein databases, which hold only what genome annotation delivers. We measured how completely the public record holds the inositol 1,4,5-trisphosphate receptor genes (*ITPR1*, *ITPR2*, *ITPR3*) that 309 vertebrate genome assemblies carry. The ground truth was a protein-to-genome sweep that owes nothing to annotation. The ryanodine receptors, a size-matched sister family sharing every diagnostic domain, were scored alongside as a control. Of 923 genome-by-paralogue genes demonstrated in the DNA, 744 (80.6 %) cannot be reached by any protein-database search, against 196 of 309 ryanodine receptor cells. Of 932 scorable loci in assemblies that ship a gene set, 689 are delivered as one complete model, which is no worse than the control's 734 of 942 ($q = 0.13$). The archive decides the outcome: 1,161 of 1,175 loci in RefSeq annotations are complete against 262 of 699 in submitter GenBank annotations. The gap narrows but does not close on assemblies able to carry the gene. Protein records name the family and paralogue correctly almost everywhere, but of 11,402 full-length records 3,872 carry a placeholder gene symbol and 2,395 carry none. All 15 vertebrate reference proteomes with no family protein are gene-caller failures. Two failures validated to the exon are intact, canonically spliced genes, and reads from 67 public RNA-seq runs cross 298 of the 314 junctions no annotated model spans. We release 297 coordinate-level corrections with their evidence.

Introduction

A gene that no protein database holds is a gene that gene-family studies, variant interpreters and structure predictors never see. Most comparative work starts from protein records, whether by name, by domain signature or by sequence similarity against a proteome, and every one of those routes depends on a genome annotation having delivered the gene as a coding model and a database having served its product. Automated annotation is known to fail [1], and new assemblies arrive faster than curation can follow them [2]. What is rarely available is a measurement, for one gene family across a declared set of genomes, of how often the record fails, in which way, and whether the failure belongs to the family or to the archive.

The inositol 1,4,5-trisphosphate (IP₃) receptors are a demanding test of that record. They are the endoplasmic reticulum's ligand-gated calcium release channels [3], encoded in vertebrates by three paralogues, *ITPR1*, *ITPR2* and *ITPR3* [4]. Each gene is about 2,700 codons spread over some 58 exons and tens to hundreds of kilobases

of genomic sequence. Mutations in them cause ataxias and other neurological disease [5], so a missing or mis-assembled record has practical cost. The family also has a built-in control. The ryanodine receptors are the IP₃ receptors' sister family [6, 7]: similar in size, sharing every diagnostic domain [8], and annotated by the same pipelines in the same assemblies. A failure rate measured on the IP₃ receptors alone cannot be told from the general quality of vertebrate gene sets. Measured beside the ryanodine receptors, it can.

Here we ask how completely the public databases record the IP₃ receptor genes that vertebrate genomes carry. The ground truth is a protein-to-genome alignment sweep over 309 vertebrate assemblies [9], which recovers each gene from the DNA without consulting any annotation, so it can be used to audit the annotations. Every locus it places is scored against the assembly's own gene set as complete, split, fragmentary, held only by a non-coding feature, or unannotated, and the ryanodine receptors are scored the same way in the same genomes. We then ask the question from the reader's side: which demonstrated genes can be reached from a protein database at all (Fig. 1). We compare curated RefSeq annotations with submitter-deposited GenBank ones (Fig. 2) and the family with its sister (Fig. 3), and score every full-length protein record the databases hold for its family, its paralogue and its name (Fig. 4). Finally we take two failures down to the exon (Fig. 5) and ask public RNA-seq whether the junctions no annotated model spans are transcribed (Fig. 6).

The answer divides the blame differently from the way the question was posed. The family is recorded no worse than its sister, and the protein records that exist are almost always named correctly. The failure lies in which archive annotated the genome, in assembly contiguity, and above all in records that do not exist.

Results

Four in five demonstrated IP₃ receptor genes have no protein record that reaches them

We first asked the reader's question directly. For every genome and every paralogue cell in which the genomic sweep demonstrates a gene, is there a protein record, in a reference proteome or elsewhere in the protein databases, that resolves to that gene? The genome sweep demonstrates 923 IP₃ receptor genes across the three paralogue cells of the 309 genomes. **744 of them (80.6 %) are reachable by no protein-database search** (Fig. 1b). The rate is similar in all three paralogues: 257 of 309 *ITPR1* genes, 260 of 307 *ITPR2* genes and 227 of 307 *ITPR3* genes have no record that resolves to them. The ryanodine receptor control cell, asked the same question in the same genomes, is unreachable in 196 of 309. The control's rate is lower but still well above half.

The 744 are missing for four different reasons, and they have different remedies. 289 sit in species with no reference proteome at all. 186 sit in species whose only family records are fragments. In 254 the species has full-length family records, but none resolves to that paralogue, and in 15 the species has no family record of any kind. Only 179 IP₃ receptor genes are held both as DNA and as a protein record assignable to their cell. The deficit does not come from the margin species the scope was extended to include. Genomes chosen as order representatives lose 413 of 556 genes and margin species 465 of 592, counting both families.

Changing the search method would not recover them. Inside one database, the vertebrate reference proteomes, an exhaustive enumeration of the family's Pfam signatures returns 3,135 gene-scale records and a pair of profile hidden Markov models returns 3,136, of which 3,135 are shared (Fig. 1a). At gene scale the two protein-level searches find the same set. What they cannot find is a gene that no record holds. The genome sweep itself contributed 1,058 gene models from 224 genomes to the census, and of the 488 that are IP₃ receptors, 318 exist only as DNA, 167 lie inside an annotated gene that carries no family name, and 3 lie inside a gene named for a different paralogue.

A quarter of annotated loci are not delivered as one complete model

The sweep places 2,144 gene-scale loci in the 309 assemblies: 1,059 IP₃ receptor loci and the ryanodine receptor control loci beside them. 255 loci sit in assemblies that ship no gene set and are reported as unscorable rather

than as failures. Of the 932 scorable IP₃ receptor loci, **689 (73.9 %) are delivered as one complete annotated model** and 243 are not (Fig. 3a). 149 are fragmentary, 14 are split across models that each deliver a piece, 42 are held only by a feature that emits no protein, and 38 have no same-strand annotated feature at all.

The completeness bar was inherited from the sweep, which calls a locus annotated when one model covers half its coding footprint, and was validated rather than re-derived. Over 1,077 loci whose own annotation names the correct paralogue on an assembly able to carry the gene, the best single model covers a median 0.993 of the gene, and only 14 of the 1,077 fall below the bar (Extended Data Fig. 1a). Moving the bar from 0.3 to 0.95 changes the complete count from 712 to 635 of 932 (Extended Data Fig. 1b), so no conclusion here depends on where it sits.

Assembly contiguity is the largest single determinant. On contigs long enough to hold the gene, the failure rate falls to 38 of 568 loci (6.7 %). Two thirds of what looks like an annotation problem is therefore an assembly too fragmented to carry a 2,700-codon gene. By vertebrate class, cartilaginous fish and mammals are nearly complete, and birds are the worst recorded: 243 of 433 bird loci are complete and 124 are fragmentary.

The family is recorded no worse than its sister family

A 58-exon gene with a sister family sharing its diagnostic domains might be expected to be annotated badly. The control shows that it is not. In the same assemblies, through the same pipelines, the ryanodine receptors are complete at 734 of 942 loci and fail at 208 (Fig. 3b). The difference from the IP₃ receptors does not survive correction for multiple testing ($q = 0.13$), and on contigs able to carry the gene the two fail at 38 of 568 and 35 of 632 loci ($q = 0.82$). How this family is recorded is how vertebrate genes of this size are recorded.

One state does separate the families, and it is the one no search can see. An IP₃ receptor locus is held only by a non-coding feature in 42 cases against 16 for the ryanodine receptors ($q = 0.006$). Such a gene is identified, often named correctly, and filed under a biotype that emits no protein, so no name- or sequence-based protein search reaches it. The excess is confined to broken assemblies: on contigs able to carry the gene the comparison is 4 against 5 ($q = 1$). The measurement cannot say whether the excess is a fact about the family or about the assemblies its loci happen to sit in.

RefSeq annotations deliver the gene whole far more often than submitter annotations do

Curated and submitter-deposited gene sets are not comparable evidence [10]. Across all scorable loci of both families, **1,161 of 1,175 loci in RefSeq annotations are complete, against 262 of 699 in GenBank annotations deposited by the submitter** (Fig. 2a). Every failure state is concentrated in the GenBank sets: 263 of their loci are fragmentary and 92 are unannotated, against 1 and 10 in RefSeq.

Submitter assemblies are also less contiguous, so the contrast was run again on loci whose contig can hold the gene (Fig. 2b). It narrows and does not close: 1,009 of 1,015 RefSeq loci are complete against 118 of 185 GenBank loci. Roughly two fifths of the gap between the archives is assembly quality, and the rest is the gene set.

Protein records are named correctly where they exist but half carry no usable symbol

We scored each of the 11,402 full-length family protein records the databases hold against a labelled panel of 38 bait proteins by blastp. Each record was assigned to a family only on a stated relative margin, and to a paralogue only within the winning family. Its own gene symbol and protein name were then read through the same verdict rule applied to the genome annotations.

The family call is not in dispute. Sequence disagrees with the census family call on 4 of 11,402 records, and 5 records carry a name from the sister family. All of these are non-vertebrate records whose best family score

is under 200 bits, where the sequence barely separates the families either. **Nor is the paralogue call.** 52 of 8,306 vertebrate records carry a symbol naming a paralogue the panel assigns elsewhere, and a further 74 name a paralogue the panel carries no bait for. Those 74 are reported as outside the instrument's reach, not as errors.

What is in dispute is whether a record can be found by name. 3,872 records carry a placeholder gene symbol and 2,395 carry none, so 55.0 % of the family's full-length records have no usable gene symbol (Fig. 4a). A further 66 protein names claim the superfamily in a form that names both families and separates neither. A search by the family's defining signature does better: PF08709 [11] is carried by 95.6 % of the 9,841 records both of this project's protein instruments call family, and by 81.2 % of the 1,554 that only one instrument found.

Every empty reference proteome is a gene-caller failure

Of 763 vertebrate reference proteomes swept with the family profiles, 15 returned no family protein at all. On its own an empty proteome cannot distinguish a species that lacks the gene from a gene caller that missed it. Each was therefore resolved against an assembly of its own species, searched with the genomic sweep. **All 15 are gene-caller failures** (Fig. 4b). Every one of the 15 species has an assembly in scope, and in every one the sweep recovers at least one IP₃ receptor locus at more than half the bait's length while the proteome holds none. The verdict vocabulary includes an *undecidable* value for a species with no assembly in scope, and it fired on none of the 15.

Two annotation failures are intact genes at the exon

A survey of states shows how often the record fails; proving that a particular annotation is wrong about a particular gene takes molecular evidence. The cases were chosen by rule. The sweep recovered 880 IP₃ receptor loci, and five eligibility rules admit the 382 that an annotation could reasonably have been expected to deliver (Extended Data Fig. 2a). The decisive rule requires the same annotation to build at least ten longer genes elsewhere in the same genome. Without it the top of the ranking consists of loci in gene sets with a genome-wide length ceiling, and the rule removes 6 loci in 2 genomes and nothing else. At 375 of the 382 eligible loci the annotation delivers the gene, and 360 have an annotation loss of exactly zero. We took the worst omission and the worst fragmentation, one per genome, to the exon (Fig. 5).

The *Nibea albiflora* *ITPR2* gene has no annotated model at all. It is a 52.5 kb locus with 56 coding exons totalling 8,021 bp, in a chromosome-level assembly whose contig N50 is 84.24 times the locus span. The annotation places no gene model over any exon, while annotating named genes 4.2 kb downstream and 5.5 kb upstream. The locus splices into 2,673 codons with no internal stop, against 14.2 stops expected under neutral drift to its observed divergence from the bait. All 55 introns carry a spliceable dinucleotide (54 GT–AG, 1 minor), and 52 of 55 exon boundaries are shared by a majority of 35 independently annotated genomes carrying the same paralogue (Extended Data Fig. 3). The gene sits beside *SSPN*, found next to this paralogue in 197 of 215 swept vertebrates. The same annotation files 7,380 of its 23,345 genes as pseudogenes, and it names its *ITPR3* locus *ITPR2* (Extended Data Fig. 2b).

The *Dissostichus eleginoides* *ITPR3* gene is annotated as three pieces. Its 60 coding exons totalling 8,064 bp are covered by 3 gene models that deliver 4,664 bp between them, leaving 3,400 bp in 28 blocks with no coding model. The locus gives 2,687 codons with no internal stop, against 11.4 expected, and all 59 introns are GT–AG. 59 of 59 exon boundaries are shared by a majority of 40 independently annotated genomes. Neither species holds any IP₃ receptor protein record, although each genome carries three complete IP₃ receptor genes.

Transcript deposits could not settle whether these genes are expressed. Probes across every junction the annotation does not model found nothing, but the two species hold only 43 and 10 transcript records. A negative control shows that the probes can fire: against the locus's own genomic DNA they return 112 and 50 hits, none spanning a junction.

Reads cross the junctions that no annotated model spans

Reads can answer the question that transcript records cannot (Extended Data Fig. 4). The panel is every locus that passes the eligibility rules with more than half its coding footprint unannotated, in species with public RNA-seq: all seven of the annotation failures, in three species, with every other family locus of the same genomes as internal controls. We aligned 67 public RNA-seq runs, 536,000,000 reads from 16 studies, to a per-species reference of spliced genomic coding sequence. **All 7 lost loci are transcribed:** each meets all three detection criteria, including at least two junction-spanning reads and more reads than its own reversed decoy, in 13 to 31 runs. Read coverage outside the annotated coding blocks tracks the exon-level loss locus by locus (Extended Data Fig. 5).

The claim that answers the annotation is per junction (Fig. 6). Of the 314 junctions in these genes that no annotated model spans, 298 (94.9 %) are crossed by reads that read through them. The same genes' annotated junctions are crossed at a similar rate. We do not rest anything on that ratio, because several libraries are strongly 3' biased and the annotated junctions of a truncated gene lie at its 5' end. The claim is the absolute one: these particular junctions are spliced.

The audit ends in 297 corrections

The audit's deliverable is addressed to the archives. **297 corrections** each give the assembly, contig coordinates, current state, current name, a proposal and an archived evidence file. The corrections comprise 20 unannotated genes, 54 genes held only by a non-coding feature, 217 incomplete models, 1 wrong paralogue name, and 5 protein records named for the wrong family. Priority is assigned by rule. High priority, 52 items, needs the gene demonstrably present, the assembly able to carry it and an intact reading frame. Anything on a contig too short to hold the gene is low priority, because there the annotation's silence may be the assembly's fault. 18 items whose loci the reading-frame screen scores as lesion-rich are kept in the list with the reason, marked as withheld rather than removed.

Discussion

The public record of the IP₃ receptor genes is much worse than the genes. Four in five of the genes the vertebrate genomes demonstrably carry cannot be reached from a protein database. More than half of the full-length protein records that do exist carry no usable gene symbol. A quarter of the loci in annotated assemblies are not delivered as one complete model. None of this is a statement about the biology of the family, and the paper's central control is what makes that sayable: the ryanodine receptors, in the same assemblies and through the same pipelines, fail at a rate indistinguishable from the IP₃ receptors'. This family is therefore an unusually well-controlled probe of how vertebrate genes of this size and structure are recorded, and the answer is general.

Three things decide whether a gene reaches the record, and none of them is the gene. The first is whether a reference proteome exists for the species: 289 of the unreachable IP₃ receptor genes are in species without one. The second is which archive annotated the genome. Curated RefSeq annotations deliver nearly every locus whole, while submitter-deposited annotations deliver about a third, and assembly contiguity explains less than half of that difference. The third is contiguity itself, which removes two thirds of the apparent annotation failures once loci on contigs too short to hold the gene are set aside. A study that samples a gene family from protein databases therefore samples annotation effort. A survey of this family built that way would have lost most of the birds, whose assemblies are fragmented and whose gene sets are largely submitter deposits.

The failures that remain on good assemblies are real failures of annotation, not of assembly or biology. The two cases taken to the exon are in chromosome-level assemblies with contigs 60 to 85 times the length of the gene. Their reading frames are open, their introns canonical, and their exon boundaries are the ones dozens of other genomes' annotations draw for the same paralogue. Reads cross almost every junction no model spans. One

of the two annotations files nearly a third of its genes as pseudogenes, so its silence at a locus that splices into an uninterrupted 2,673-codon reading frame looks like a pipeline setting, not a judgement about this gene. The companion analysis of gene architecture reaches the same conclusion from the other side: most fragmentary models of this family stop inside an exon rather than at a junction, which is not where a real gene boundary can lie (companion paper 2).

The protein-record findings point to a cheap remedy. The databases are not confusing the families or the paralogues; the sequence and the name disagree on a handful of records out of 11,402. What they lack is symbols. A record that carries a placeholder or no symbol is invisible to a name-based query and legible only to a sequence search. Assigning symbols to records whose sequence already places them would recover more findability than reannotating any genome.

The audit has limits that bound what it can claim. Its ground truth is one instrument, a protein-to-genome sweep, and a locus that instrument did not find cannot appear here as an annotation failure. The sweep misses genes in fragmented assemblies (companion paper 3), so the failure rates reported are, if anything, conservative on poor assemblies. The completeness bar was inherited rather than chosen, but it was validated against the population it acts on, and every state is re-counted across the whole plausible range. The excess of IP₃ receptor loci held only by non-coding features is real in the record set, yet it disappears on assemblies able to carry the gene. We therefore do not claim it as a property of the family. The expression evidence covers the seven failures in three fish species that have public RNA-seq. It shows that those particular unannotated junctions are spliced, not that every unannotated locus is transcribed. The 297 corrections are proposals from one instrument about genes that instrument recovered; two are validated to the exon and by reads, and the rest rest on the same instrument applied at scale.

The deliverable is addressed to somebody else. Each correction carries coordinates, the current state, a proposal and an evidence file a curator can open, and the priority rule says which ones a curator can act on without re-examining the assembly.

Methods

Scope

The genome scope is 309 vertebrate assemblies from NCBI Datasets [10], declared before any search as the best assembly of every vertebrate order, ranked by annotation status, RefSeq category, assembly level and scaffold N50, together with margin species chosen by stated rules from the protein census. 168 of the assemblies carry a RefSeq annotation and 141 a GenBank annotation or none; every locus is scored against the gene set that ships with its own assembly and against no other. The genes audited are the sweep's gene-scale loci, 2,144 in all, of which 1,059 are IP₃ receptor loci. Loci in assemblies that ship no gene set are counted and reported as unscorable, leaving 932 scorable IP₃ receptor loci and 942 scorable ryanodine receptor loci. The protein scope is the 11,402 full-length family records (at least 2,000 residues, not flagged as fragments, from a protein database rather than a genome model), of which 8,306 are vertebrate. The reference-proteome scope is the 763 vertebrate UniProt reference proteomes [12].

Contiguity is carried as a stratum everywhere. A locus is on a contig able to carry the gene when that contig is at least as long as the median measured IP₃ receptor gene span, 142,212 bp. Every comparison is reported over all scorable loci and again over loci that clear this bar, because a contig shorter than the gene cannot be annotated completely by anybody.

The ground truth

Each assembly was searched with miniprot [9] using a panel of 38 bait proteins, 30 IP₃ receptors across the vertebrate clades and 8 ryanodine receptors as a presence control. Loci were clustered, called to family on

alignment score before any paralogue was considered, and assigned to a paralogue cell inside the winning family. Empty cells were re-searched by whole-genome tblastn [13]. The sweep reads no annotation when it places a gene, which is what allows it to serve as an independent reference for the annotation. Each locus carries its coding blocks in gene order, its bait coverage and identity, and its frameshift and stop counts.

Locus states

Every scorable locus was assigned one of five states against the assembly's own GFF3 annotation: *complete* (one coding model covers at least half the locus's aligned coding footprint), *split* (two or more coding models each contribute a piece and none reaches the bar), *fragmentary* (the coding models present deliver less than the bar), *non-coding only* (a same-strand feature covers the locus but emits no protein, most often a pseudogene) and *unannotated* (no same-strand feature). Three rules make the state a measurement. Only same-strand features count, because an antisense gene is not a model of this one. Loci are scored against coding blocks and never gene spans, because these genes' introns reach 152 kb and a passenger gene's span inside one would otherwise cover the locus. The name is read from whichever model the annotation places over the locus, preferring a small correctly named model to a large unnamed one, because accusing a database of failing to name a gene it did name is the costlier error.

The name verdict has two values that exist because their absence manufactured errors. A name that claims the family but no paralogue is recorded as *paralogue unspecified*, not as wrong. A name that claims the superfamily in a form naming both families is recorded as *family ambiguous*, not as a wrong-family call.

Controls

The sister family is the central control. Every ryanodine receptor locus the sweep placed was scored through the same states, names and bars in the same assemblies. Family and control were compared state by state with Fisher's exact test and Benjamini–Hochberg correction [14] across all states within each scope. This is the comparison that separates a statement about this family from a statement about vertebrate gene sets. Its outcome is reported in the Results: no overall difference in either scope.

The completeness bar was validated, not chosen. It was inherited from the sweep, where it decides whether a found gene counts as annotated, and was validated on 1,077 loci whose own annotation names the correct paralogue, whose alignment recovers at least 90 % of the bait, and whose contig clears the contiguity bar. The bar sits at that distribution's 1.3 % point. Every state was re-counted at bars from 0.3 to 0.95.

The source contrast was run twice, over all scorable loci and again over loci that clear the contiguity bar, with the same test and correction, so that the archive effect and the assembly effect are reported side by side.

Protein records and proteomes

Each full-length record was compared by blastp against the bait panel, assigned to a family only when its best score in one family beats the other by more than a relative margin of 0.1, and to a paralogue within the winning family by the same rule. The family question was asked of every record and the paralogue question of vertebrate records only. A record whose symbol names a paralogue the panel carries no bait for is reported as outside the instrument's reach. A protein-record rename was proposed only when the record's best score also cleared 200 bits. Each reference proteome with no family hit in the profile sweep [15] was resolved against an assembly of its own species under four verdicts: gene caller missed it, genome also empty, assembly cannot carry the gene, and undecidable because no assembly is in scope.

Reachability

For each genome and each paralogue cell in which the sweep demonstrates a gene (1,232 genome-by-cell genes, 923 of them IP₃ receptors), a gene counts as reachable when at least one full-length protein record of the same species resolves to that cell. Unreachable genes are sorted by reason: no reference proteome for the species, only fragmentary records, records present but none resolving to the cell, or no family record at all. The same classification was applied to the ryanodine receptor cell. The protein-level search channels were compared inside the vertebrate reference proteomes, counting only records each channel calls family and only accessions the swept files hold.

Case selection and validation

Annotation loss is the fraction of a recovered gene's coding footprint that no single annotated model delivers. It was ranked over every recovered IP₃ receptor locus that passes five eligibility rules: the assembly carries a gene set; the sweep recovered the gene at coverage of at least 0.7 with no contig edge or gap; the contig holds the locus; contig N50 is at least ten times the locus span; and at least ten annotated protein-coding genes elsewhere in the same genome are longer than the locus. The fifth rule is the per-genome annotation-depth control; it removes 6 loci in 2 genomes. Failures were labelled omission, truncation or fragmentation, and the worst of two modes was taken, one case per genome, with ties broken by how many other family loci in the same genome the annotation gets right.

For each case the spliced coding sequence was reconstructed from the aligned blocks and its internal stops compared with the number expected under neutral drift to the observed divergence, computed from the locus's own codon usage. The comparison is one-sided: zero stops falsifies a pseudogene call, and a handful would not establish one. Splice dinucleotides were read from the genome with the strand corrected. **Exon-boundary concordance** is the control on the instrument: every swept genome whose same paralogue was aligned from the same bait, and whose own annotation independently places one model over the whole alignment, contributes its boundaries, and each case boundary is scored by the fraction of those genomes sharing it. Annotated proteins were translated from each assembly's own annotation and tiled onto the genome's own recovered loci by blastp. Junction probes of 90 nucleotides either side were searched against every transcript record of the species, counting only hits contiguous across the junction with at least 8 nucleotides on each side. The same probes were searched against the locus's own genomic DNA as a negative control that cannot span a junction.

Expression

The expression panel took every eligible locus with annotation loss above 0.5 in species with at least four public Illumina RNA-seq runs, plus every other family locus of the same genomes as internal controls. Runs were chosen from the Sequence Read Archive [16] by tissue keyword, verified against each run's own sample attributes, spread across studies, and streamed through HISAT2 [17] with spliced alignment disabled, because the reference is already coding sequence. Each reference sequence is the spliced genomic coding sequence of its locus, deliberately not corrected for frameshifts, validated by placing every block colinearly in the sweep's own protein. Three housekeeping anchors were built the same way.

Two controls make the counts readable. Every reference carries a reversed decoy of identical length and composition, and a locus is detected in a run only when it has at least two junction-spanning reads, at least five reads, and more reads than its own decoy. Across all runs the decoys collected 42 reads in 2 of 469 run-by-locus comparisons, and every affected locus still cleared its decoy. Cross-mapping between paralogues was measured rather than assumed: every reference was tiled with 100-nucleotide synthetic reads at 10-nucleotide steps and mapped back under the same settings, and 0 of 12,500 reads were assigned to a sequence other than their source. A read spans a junction when one contiguous aligned block covers at least 8 nucleotides on both sides. Junctions

with no reads are written as zeros, not omitted.

Corrections

A correction is written for every scorable locus not delivered complete and for every protein record whose name and sequence disagree on family. Priority is high when the gene is demonstrably present, its contig clears the contiguity bar and its reading frame is intact; medium for a partial recovery or an unscored reading frame; and low below the contiguity bar. Where the reading-frame screen scores a locus as lesion-rich, the row is written, marked as withheld and given its reason, so the list records what the audit declined to propose.

Negative controls and reproducibility

Every stage runs a suite of constructed negative controls before writing anything: 44 for the locus and protein audit, 47 for the case validation and 11 for the expression analysis. Each suite was mutation-tested by breaking rules deliberately. Every table and figure is generated by script from committed tables, and every number in this paper is declared in a claims ledger with the table it comes from and re-verified on every build.

Figure legends

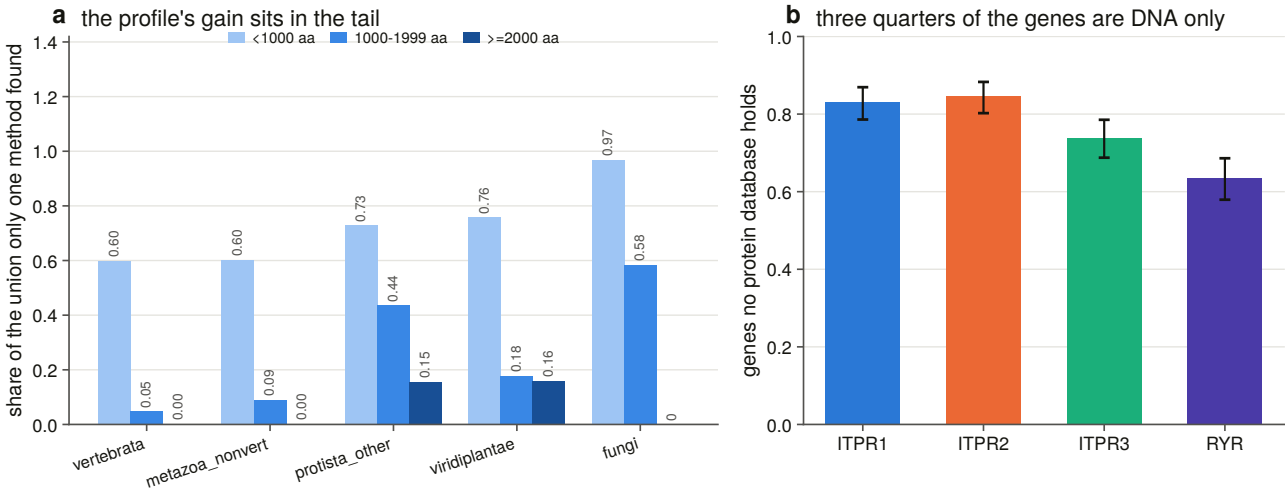


Fig. 1. Most demonstrated genes have no protein record, and a better protein search would not change that. Panel (a) compares two protein-level searches of the same reference proteomes, an exhaustive enumeration of the family's Pfam signatures and a pair of profile hidden Markov models. Each bar gives the share of the records either search calls family that only one of them found, split by record length, for each swept group. The profile adds nothing at gene scale in the vertebrates or the other animals, so its gain lies in the fragment tail. Panel (b) asks, for each paralogue cell and for the ryanodine receptor control cell, what fraction of the genes the genome sweep demonstrates no protein record resolves to, with 95 % Wilson intervals. The whiskers do not reach below 0.68 for any IP₃ receptor paralogue.

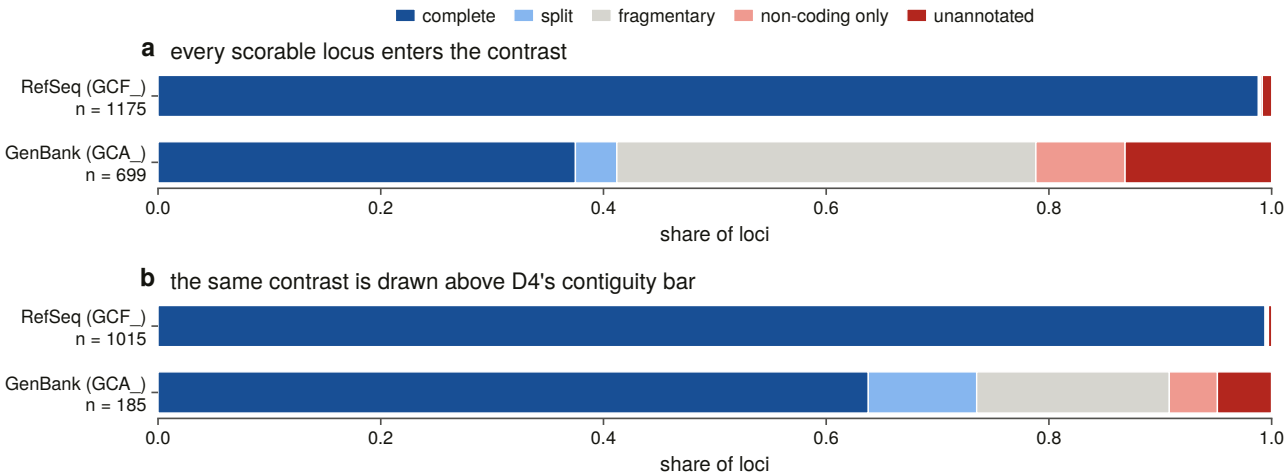


Fig. 2. Loci in RefSeq annotations are delivered whole far more often than loci in submitter annotations. Panel (a) stacks the five locus states for every scorable locus, split by whether the assembly's gene set is a curated RefSeq annotation or a GenBank annotation deposited by the submitter. Panel (b) repeats the contrast on the loci whose contig is long enough to hold the gene. The gap narrows in (b) without closing, which separates the archive effect from the assembly effect: GenBank assemblies are less contiguous, yet the GenBank gene sets still fail on good contigs.

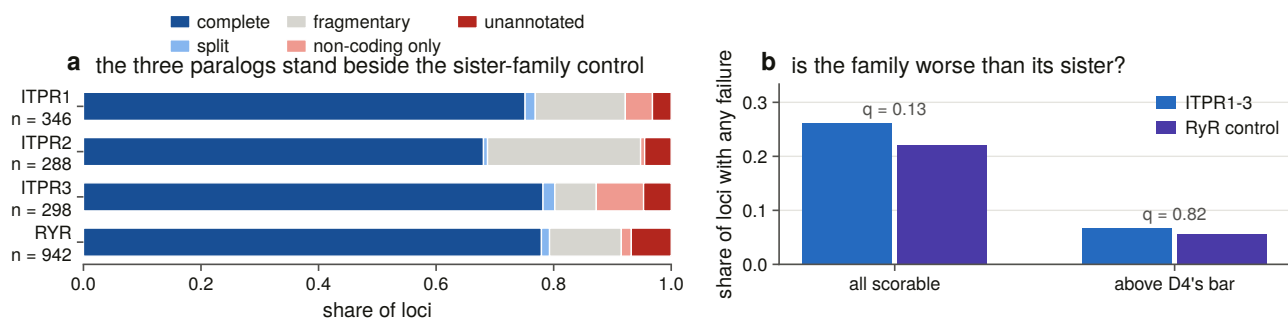


Fig. 3. The IP_3 receptors are recorded at the rate their sister family is recorded in the same assemblies. Panel (a) stacks the five locus states for each of *ITPR1*, *ITPR2* and *ITPR3* beside the ryanodine receptor control. Panel (b) compares the share of loci with any annotation failure between the family and the control over all scorable loci and over loci on contigs able to carry the gene, with the Benjamini–Hochberg corrected p-value above each pair. Neither pair differs after correction, so a quarter of loci missing a complete model is the rate for vertebrate genes of this size, not a defect peculiar to this family.

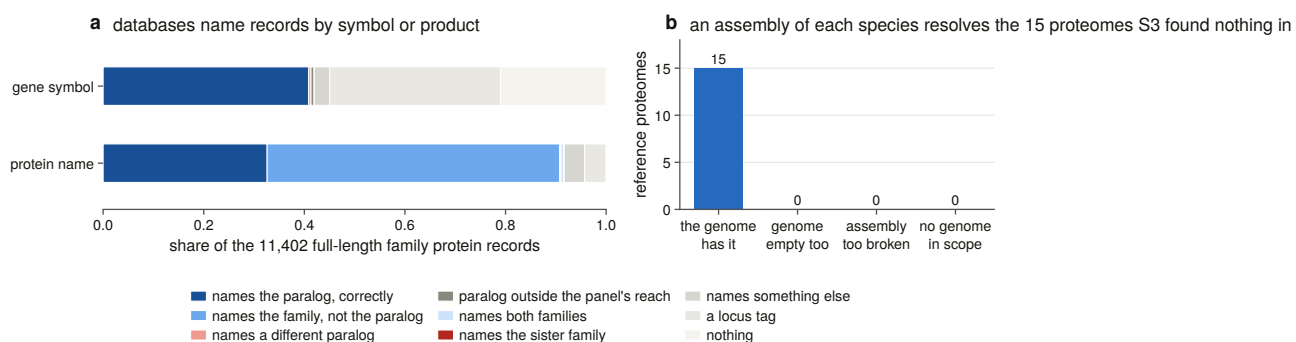


Fig. 4. Protein records that carry a name are almost always named correctly, and most carry none. In panel (a), each of the 11,402 full-length family records is scored twice, on its gene symbol and on its protein name, against the paralogue its sequence is assigned to. The categories run from a correct paralogue, through the family named without a paralogue and names outside the panel's reach, to a wrong paralogue, the sister family, both families at once, a locus tag, or nothing. The pale bands of placeholder and absent symbols fill more than half the symbol bar. Panel (b) sorts the 15 vertebrate reference proteomes with no family hit by what an assembly of the same species holds, and every one falls where the genome holds the gene and the proteome does not.

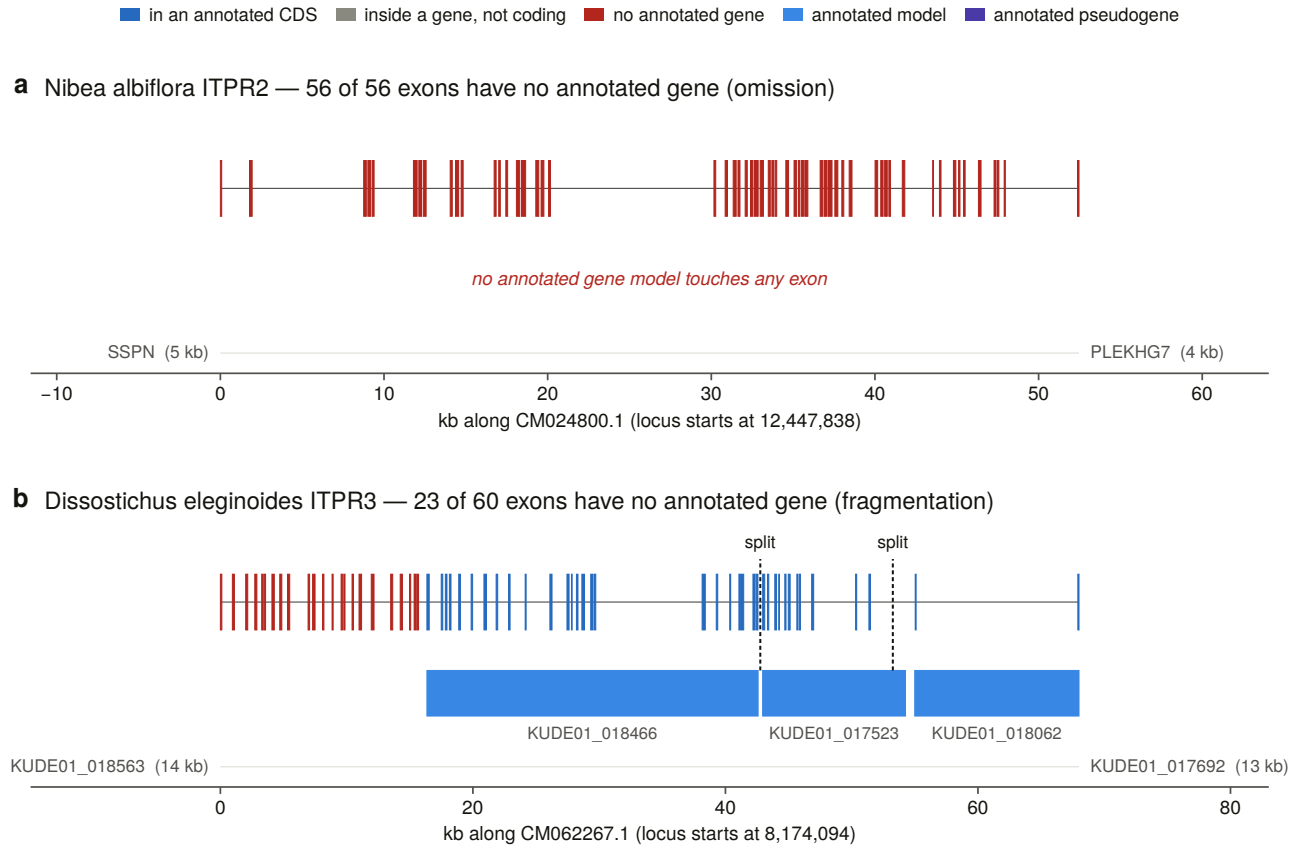


Fig. 5. Two annotation failures are drawn at true genomic width, each exon at its real size. Panel (a) is the *Nibea albiflora* ITPR2 locus, where all 56 exons are coloured as touched by no annotated gene. Panel (b) is the *Dissostichus eleginoides* ITPR3 locus, with the three annotated models drawn beneath the exons they cover and 23 of the 60 exons outside all of them. Flanking named genes mark each end. Exons are never widened for visibility, because a 56-exon gene over 52 kb averages 143 bp an exon and a widened drawing would misstate how little of the locus the missed sequence occupies.

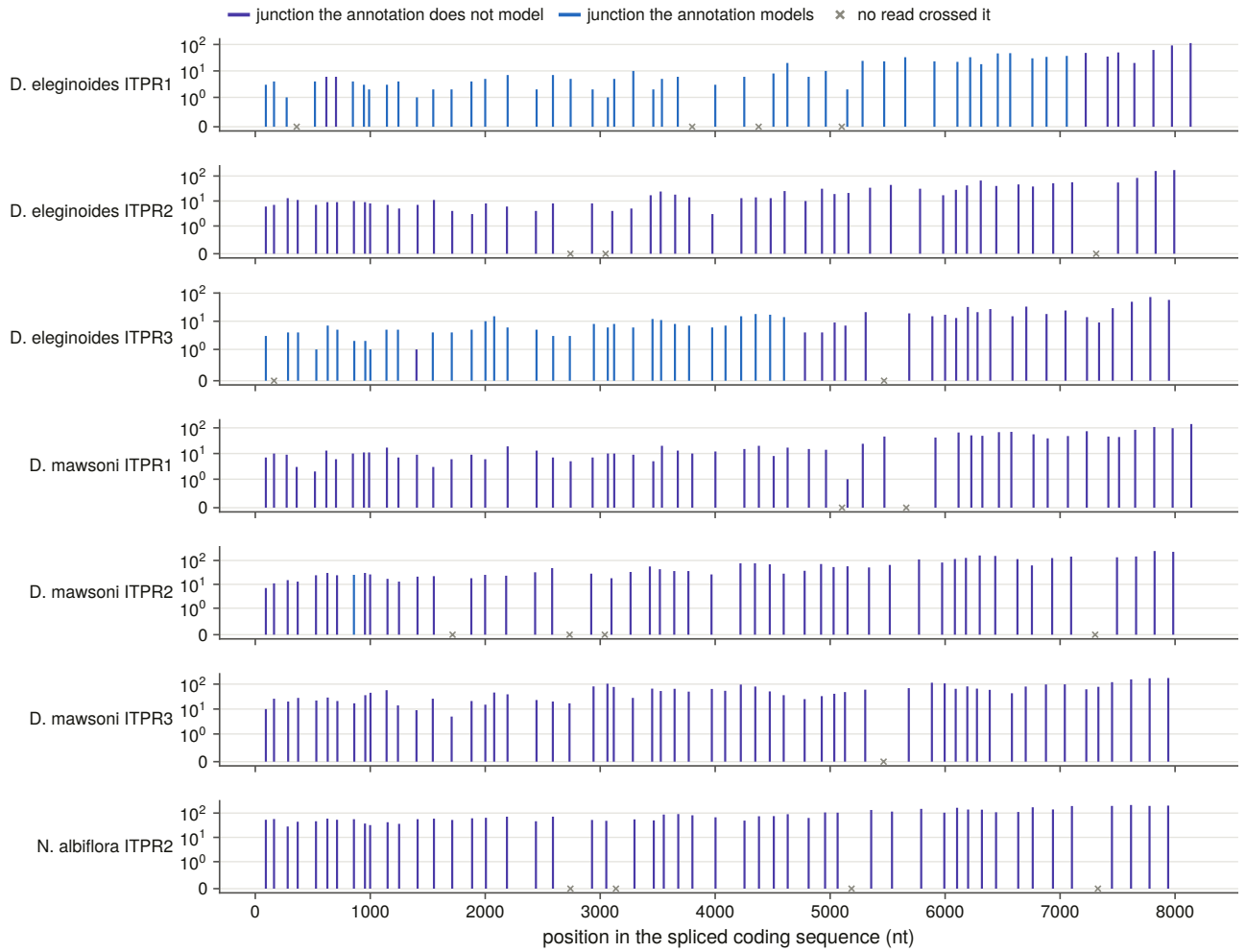
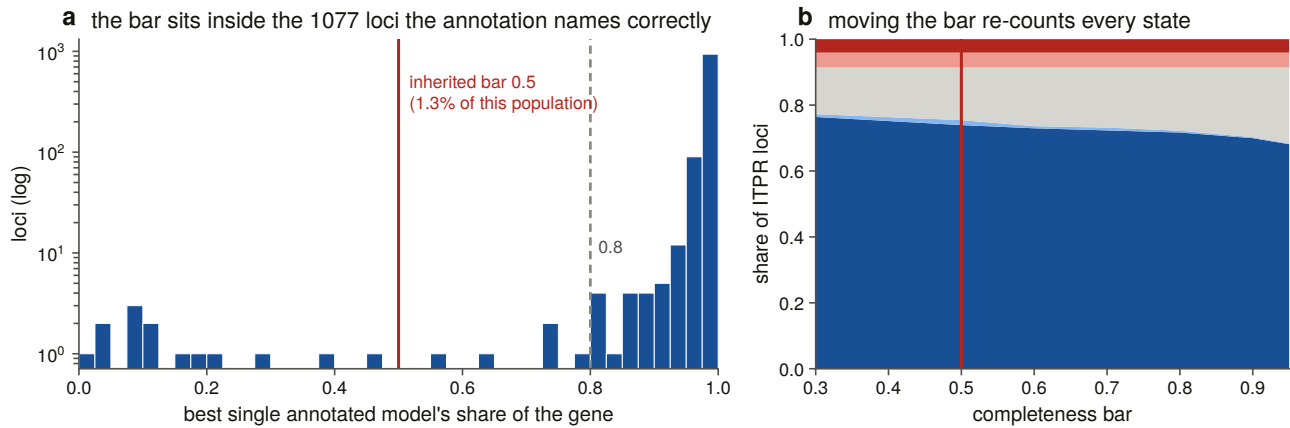
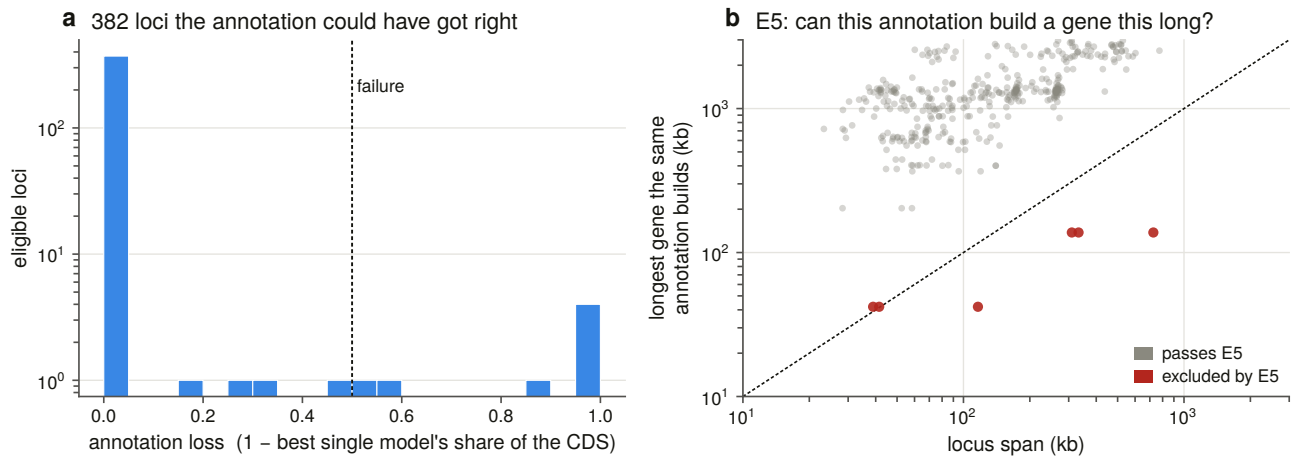


Fig. 6. Reads cross the junctions that no annotated model spans. Each row is one of the seven IP₃ receptor genes the annotation loses, and each junction of its spliced coding sequence is a bar at its position along the gene, with height equal to the number of reads that read through it. Colour marks whether an annotated model contains the junction, and a cross on the baseline marks a junction no read crossed, so the denominator is visible. Transcription alone would say little about a partly annotated gene; spliced molecules across the missing introns show that the unmodelled exon structure exists.

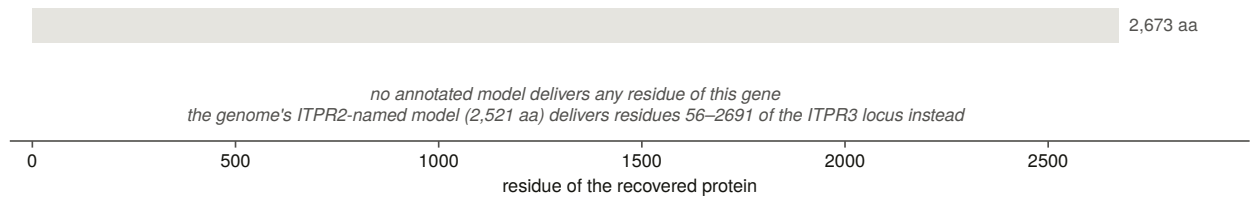
Extended Data



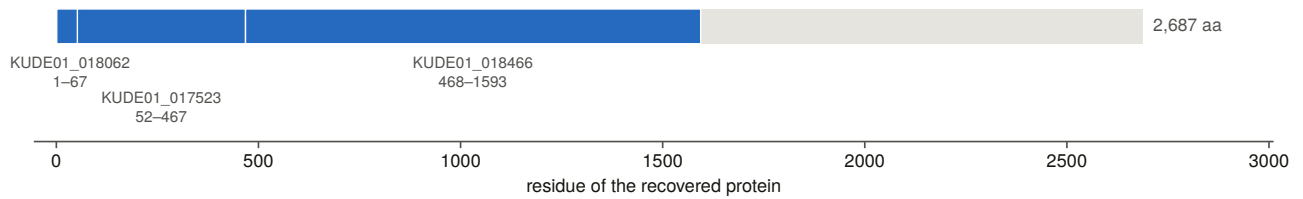
Extended Data Fig. 1. The completeness bar sits in the thin tail of the population it was validated against. Panel a is the share of the gene that the best single annotated model delivers, on a logarithmic count axis, over the 1,077 loci whose own annotation names the correct paralogue on an assembly able to carry the gene. The inherited bar at 0.5 is drawn where it falls, below 1.3 % of the population. Panel b re-counts every IP_3 receptor locus state as the bar moves from 0.3 to 0.95, and the complete band narrows only slightly across the whole range. A number in a legend cannot show whether a threshold falls in a gap or in the middle of a population, and this panel shows that it falls in the gap.



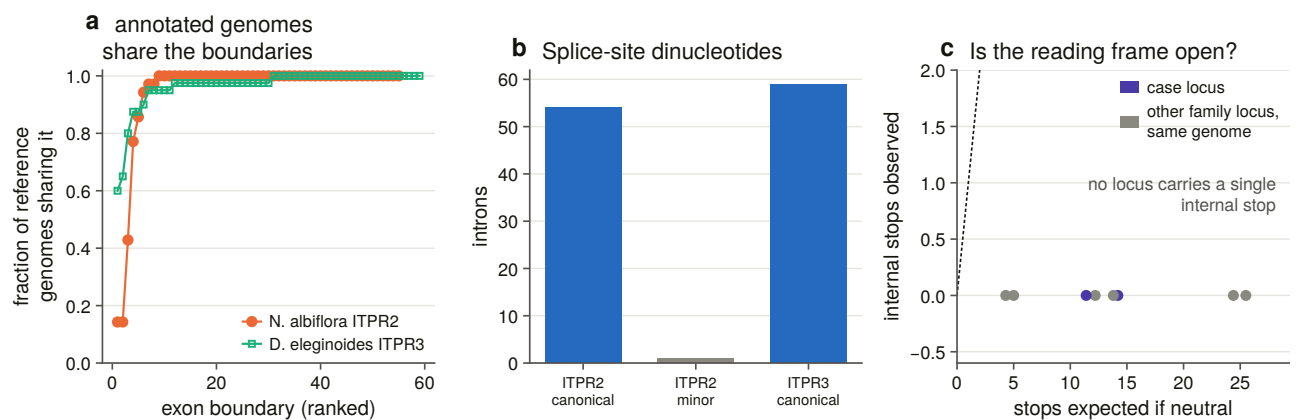
a *Nibea albiflora* ITPR2 — annotated models deliver 0 of 2,673 residues (0%)



b *Dissostichus eleginoides* ITPR3 — annotated models deliver 1,609 of 2,687 residues (60%)

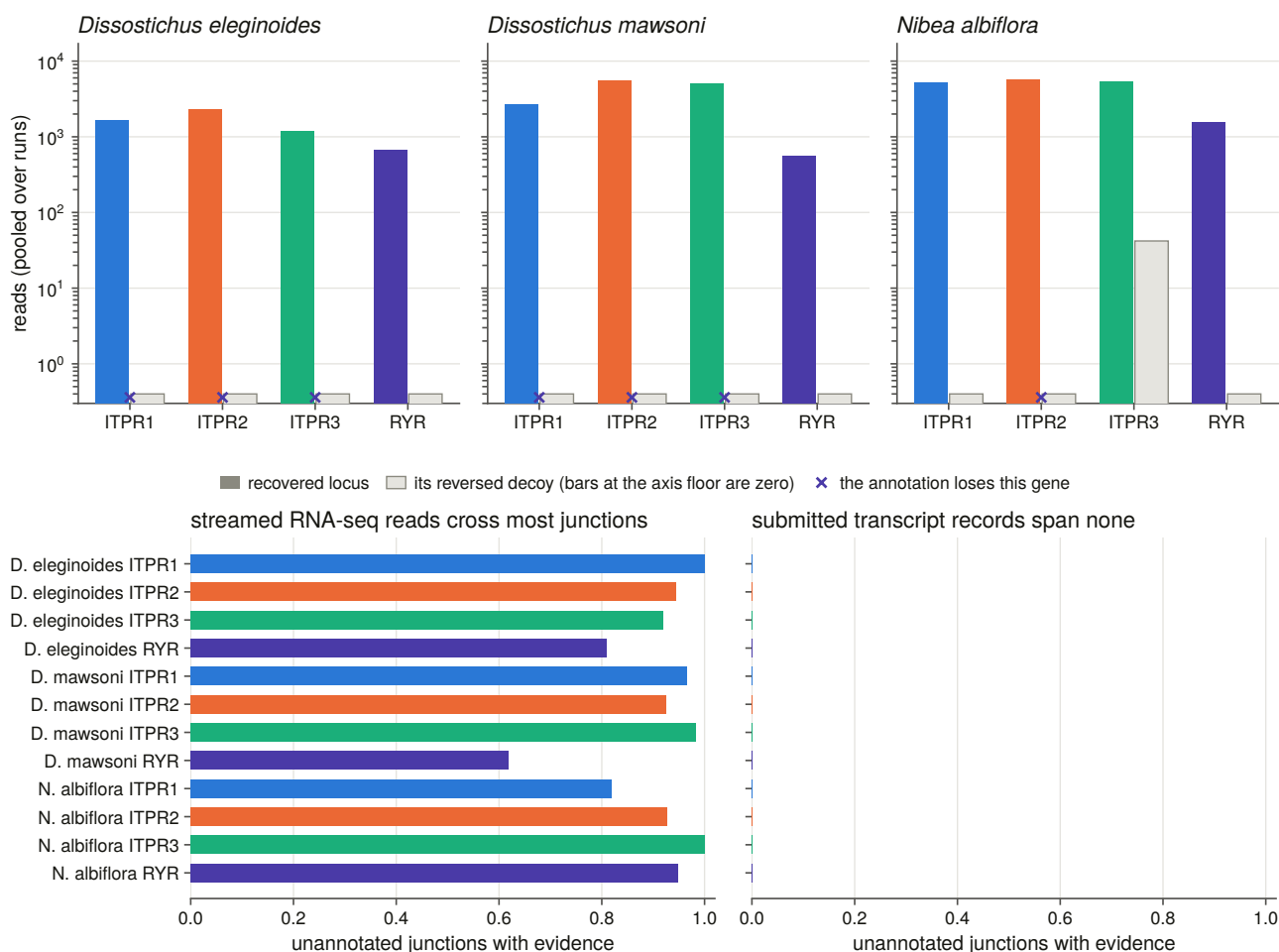


Extended Data Fig. 2. The validated cases come from a measured tail, and the annotation's own proteins show what it delivers at each. (a) Annotation loss over the 382 eligible loci has 360 at zero, on a logarithmic count axis. Beside it, the fifth eligibility rule plots each locus's span against the longest gene its own annotation builds and marks in red the six loci it excludes. (b) The annotation's own proteins, translated from its GFF3 and genome, are tiled onto each case's recovered protein. No model delivers any residue of the *Nibea albiflora* ITPR2 gene, and the model the annotation names ITPR2 tiles onto the ITPR3 locus instead. Three models deliver 1,609 of the 2,687 residues of the *Dissostichus eleginoides* ITPR3 gene.

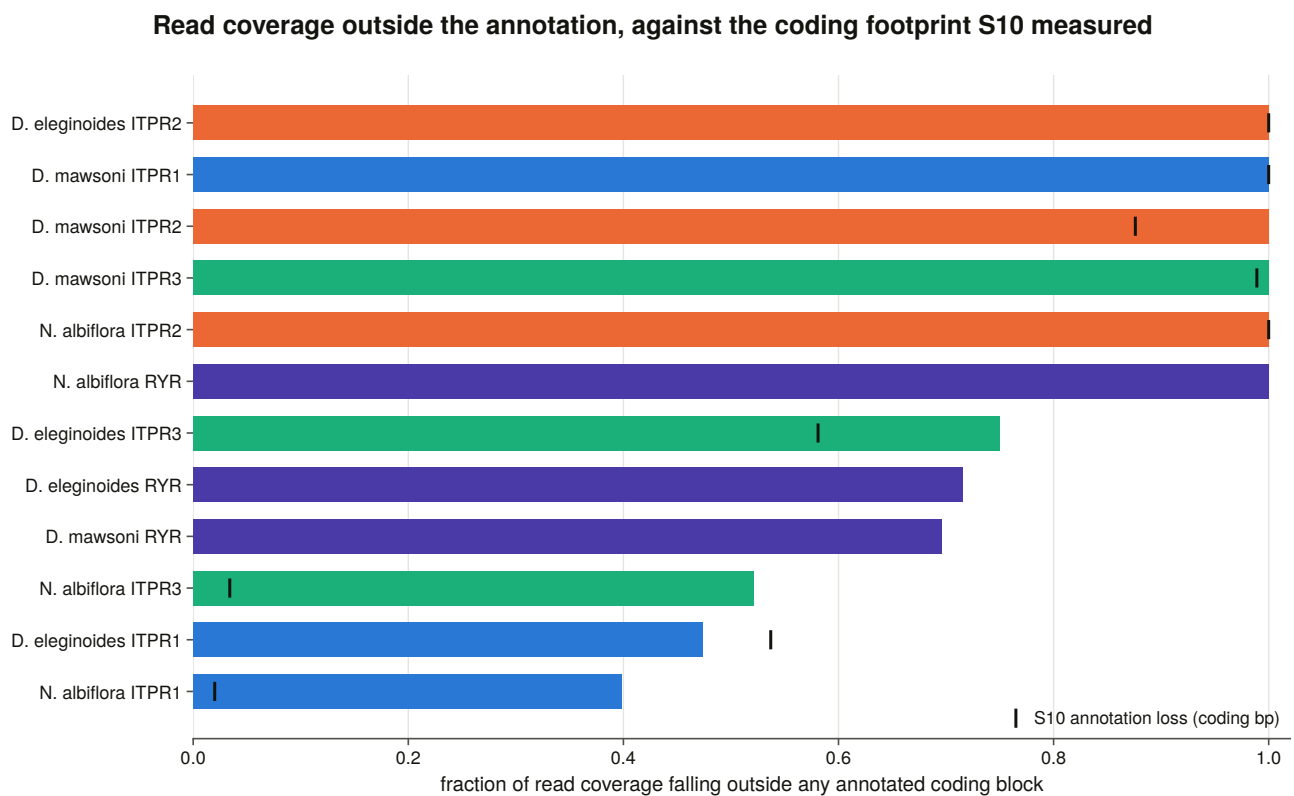


Extended Data Fig. 3. Three independent checks agree on both cases. Panel a ranks the exon boundaries of each case by the share of independently annotated genomes that draw them, and that share is above 0.9 within the first ten boundaries of both loci. Panel b counts the introns of each case by splice dinucleotide, and all but one of them have a canonical pair. In panel c, no case locus has an internal stop, and neither does any other family locus of the same genomes, against 4 to 25 stops expected under neutral drift. No single check is decisive, but they fail in different circumstances and none fails here.

Each recovered locus collects reads, scored against its own composition-matched decoy



Extended Data Fig. 4. Read evidence rises far above its controls, and it answers the question the transcript deposits cannot. (a) The pooled read count on every family locus of the three species stands beside the count on its own reversed decoy on a logarithmic axis, and a cross marks each locus the annotation loses. Every decoy but one sits at the floor, and the exception collects a few dozen reads against several thousand on its locus. (b) Streamed reads cross most unannotated junctions at every locus, while the species' submitted transcript records cross none. A species with ten transcript deposits cannot show that a gene is silent, and reporting that search as underpowered keeps a database's emptiness from reading as an absence.



Extended Data Fig. 5. Two independent instruments agree about the sequence the annotation loses. Each bar is the fraction of one locus's read coverage that falls outside any annotated coding block, and the tick on the bar is the share of the coding footprint that the exon-level measurement found no model delivering. One instrument aligns a protein to a genome and the other maps reads to a spliced reference, so their agreement rules out the loss being an artefact of either.

Data availability

The 297 corrections are offered as a table, `results/annotation_audit/corrections.tsv`, one row per proposal with the assembly accession, the annotation source, the contig coordinates and strand, the current state, name and biotype, the proposal, the priority with the rule that assigned it, the withheld flag with its reason, and the path of an archived evidence file a curator can open. The table is deposited with this paper and is intended for submission to the RefSeq and GenBank annotation teams and to UniProt.

Every committed table and figure behind this paper is listed file by file, with its size and SHA-256, in `papers/archive/deposit_manifest.tsv`, and `papers/archive/deposit_notes.md` names each class of bulk data excluded with the command that regenerates it. All primary data are public and were used unmodified: NCBI Datasets genome assemblies and their annotations for the 309 genomes of the scope, UniProt reference proteomes and protein records, and the Sequence Read Archive runs listed in `results/expression/runs_selected.tsv`. No new sequence data were generated.

Code availability

The analysis is a single repository of scripted stages, each rendering its report from committed tables and each running constructed negative controls before it writes anything. `python scripts/s26_assemble.py --paper archive` rebuilds this paper from those tables and exits non-zero on a missing figure, a cited reference with no bibliography row, or a load-bearing number that no longer matches its source table.

Author contributions

George Dickinson conceived and directed the project and checked its results. The analysis code and the text of this paper were written by Claude (Anthropic) under his direction; Chapter 16 of the accompanying thesis describes how the work was divided and what was corrected.

Competing interests

The author declares no competing interests.

Companion papers

2. George Dickinson. The three vertebrate IP₃ receptors arose in two separate early vertebrate duplications. *In preparation* (paper 2 of 6 in this series; source in `papers/origin/`).
3. George Dickinson. No vertebrate has lost an IP₃ receptor paralogue. *In preparation* (paper 3 of 6 in this series; source in `papers/retention/`).

References

1. Salzberg SL. Next-generation genome annotation: we still struggle to get it right. *Genome Biol* 2019. doi:[10.1186/s13059-019-1715-2](https://doi.org/10.1186/s13059-019-1715-2). PMID 31097009. [R194]
2. Rhie A, McCarthy SA, Fedrigo O, Damas J, Formenti G, Koren S, Uliano-Silva M, Chow W, Fungtammasan A, Kim J, Lee C, Ko BJ, Chaisson M, Gedman GL, Cantin LJ, Thibaud-Nissen F, Haggerty L, Bista I, Smith M, Haase B, Mountcastle J, Winkler S, Paez S, Howard J, Vernes SC, Lama TM, Grutzner F, Warren WC, Balakrishnan CN, Burt D, George JM, Biegler MT, Iorns D, Digby A, Eason D, Robertson B, Edwards T, Wilkinson M, Turner G, Meyer A, Kautt AF, Franchini P, Detrich HW 3rd, Svardal H, Wagner M, Naylor GJP, Pippel M, Malinsky M, Mooney M, Simbirsky M, Hannigan BT, Pesout T, Houck M, Misuraca A, Kingan SB, Hall R, Kronenberg Z, Sović I, Dunn C, Ning Z, Hastie A, Lee J,

- Selvaraj S, Green RE, Putnam NH, Gut I, Ghurye J, Garrison E, Sims Y, Collins J, Pelan S, Torrance J, Tracey A, Wood J, Dagnew RE, Guan D, London SE, Clayton DF, Mello CV, Friedrich SR, Lovell PV, Osipova E, Al-Ajli FO, Secomandi S, Kim H, Theofanopoulou C, Hiller M, Zhou Y, Harris RS, Makova KD, Medvedev P, Hoffman J, Masterson P, Clark K, Martin F, Howe K, Flicek P, Walenz BP, Kwak W, Clawson H, Diekhans M, Nassar L, Paten B, Kraus RHS, Crawford AJ, Gilbert MTP, Zhang G, Venkatesh B, Murphy RW, Koepfli KP, Shapiro B, Johnson WE, Di Palma F, Marques-Bonet T, Teeling EC, Warnow T, Graves JM, Ryder OA, Haussler D, O'Brien SJ, Korlach J, Lewin HA, Howe K, Myers EW, Durbin R, Phillippy AM, Jarvis ED. Towards complete and error-free genome assemblies of all vertebrate species. *Nature* 2021. doi:[10.1038/s41586-021-03451-0](https://doi.org/10.1038/s41586-021-03451-0). PMID 33911273. [R195]
3. Berridge MJ. Inositol trisphosphate and calcium signalling. *Nature* 1993. doi:[10.1038/361315a0](https://doi.org/10.1038/361315a0). PMID 8381210. [R07]
 4. Ross CA, Danoff SK, Schell MJ, Snyder SH, Ullrich A. Three additional inositol 1,4,5-trisphosphate receptors: molecular cloning and differential localization in brain and peripheral tissues. *Proc Natl Acad Sci USA* 1992. doi:[10.1073/pnas.89.10.4265](https://doi.org/10.1073/pnas.89.10.4265). PMID 1374893. [R32]
 5. Foskett JK. Inositol trisphosphate receptor Ca²⁺ release channels in neurological diseases. *Pflugers Arch* 2010. doi:[10.1007/s00424-010-0826-0](https://doi.org/10.1007/s00424-010-0826-0). PMID 20383523. [R122]
 6. Mignery GA, Sudhof TC, Takei K, De Camilli P. Putative receptor for inositol 1,4,5-trisphosphate similar to ryanodine receptor. *Nature* 1989. doi:[10.1038/342192a0](https://doi.org/10.1038/342192a0). PMID 2554146. [R04]
 7. Mackrill JJ. Ryanodine receptor calcium release channels: an evolutionary perspective. *Adv Exp Med Biol* 2012. doi:[10.1007/978-94-007-2888-2_7](https://doi.org/10.1007/978-94-007-2888-2_7). PMID 22453942. [R37]
 8. Seo MD, Velamakanni S, Ishiyama N, Stathopoulos PB, Rossi AM, Khan SA, Dale P, Li C, Ames JB, Ikura M, Taylor CW. Structural and functional conservation of key domains in InsP3 and ryanodine receptors. *Nature* 2012. doi:[10.1038/nature10751](https://doi.org/10.1038/nature10751). PMID 22286060. [R58]
 9. Li H. Protein-to-genome alignment with miniprot. *Bioinformatics* 2023. doi:[10.1093/bioinformatics/btad014](https://doi.org/10.1093/bioinformatics/btad014). PMID 36648328. [R164]
 10. O'Leary NA, Cox E, Holmes JB, Anderson WR, Falk R, Hem V, Tsuchiya MTN, Schuler GD, Zhang X, Torcivia J, Ketter A, Breen L, Cothran J, Bajwa H, Tinne J, Meric PA, Hlavina W, Schneider VA. Exploring and retrieving sequence and metadata for species across the tree of life with NCBI Datasets. *Sci Data* 2024. doi:[10.1038/s41597-024-03571-y](https://doi.org/10.1038/s41597-024-03571-y). PMID 38969627. [R159]
 11. Mistry J, Chuguransky S, Williams L, Qureshi M, Salazar GA, Sonnhammer ELL, Tosatto SCE, Paladin L, Raj S, Richardson LJ, Finn RD, Bateman A. Pfam: The protein families database in 2021. *Nucleic Acids Res* 2021. doi:[10.1093/nar/gkaa913](https://doi.org/10.1093/nar/gkaa913). PMID 33125078. [R155]
 12. UniProt Consortium. UniProt: the Universal Protein Knowledgebase in 2023. *Nucleic Acids Res* 2023. doi:[10.1093/nar/gkac1052](https://doi.org/10.1093/nar/gkac1052). PMID 36408920. [R157]
 13. Camacho C, Coulouris G, Avagyan V, Ma N, Papadopoulos J, Bealer K, Madden TL. BLAST+: architecture and applications. *BMC Bioinformatics* 2009. doi:[10.1186/1471-2105-10-421](https://doi.org/10.1186/1471-2105-10-421). PMID 20003500. [R158]
 14. Benjamini Y, Hochberg Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 1995. doi:[10.1111/j.2517-6161.1995.tb02031.x](https://doi.org/10.1111/j.2517-6161.1995.tb02031.x). [R170]
 15. Eddy SR. Accelerated Profile HMM Searches. *PLoS Comput Biol* 2011. doi:[10.1371/journal.pcbi.1002195](https://doi.org/10.1371/journal.pcbi.1002195). PMID 22039361. [R153]

16. Leinonen R, Sugawara H, Shumway M, International Nucleotide Sequence Database Collaboration. The sequence read archive. *Nucleic Acids Res* 2011. doi:[10.1093/nar/gkq1019](https://doi.org/10.1093/nar/gkq1019). PMID 21062823. [R163]
17. Kim D, Paggi JM, Park C, Bennett C, Salzberg SL. Graph-based genome alignment and genotyping with HISAT2 and HISAT-genotype. *Nat Biotechnol* 2019. doi:[10.1038/s41587-019-0201-4](https://doi.org/10.1038/s41587-019-0201-4). PMID 31375807. [R165]